CAP182 SS2 - Part D: Recommendation Report

To	STADIOEquities
From	Vusumuzi Nkosi (Student number 26303296)
Date	1 September 2026
Subject	Public-data proof of viability and recommended next steps

Introduction

This report assesses the viability of the proposed modelling technique for account activation as outlined in SS1 prior to the exposure of clients' information by STADIOEquities. The Logistic Regression model and the XGBoost model have been fitted on the UCI Bank Marketing public dataset. In this particular problem, we need to make predictions about the subscription to the term deposit of a client. It is not the same business occurrence as that of the initial investment deposit, however, it serves as an adequate example due to the imbalance of the binary outcome variable.

1. Recommended model

I would suggest using Model 2 (XGBoost) as the better proof-of-viability model. It showed improvements in accuracy (from 0.8339 to 0.8496), balanced accuracy (from 0.7536 to 0.7614), precision (from 0.3663 to 0.3972), F1 (from 0.4685 to 0.4924), ROC-AUC (from 0.8020 to 0.8090), and PR-AUC (from 0.4591 to 0.4868) scores on the test set of 8,238 records. Recall score increased slightly (from 0.6476 to 0.6498), meaning that this model has become better not through giving up a high number of likely subscribers.

It becomes apparent in the confusion matrices too. XGBoost detected 601 true positives against 603 found by Logistic Regression but decreased false positives from 1,043 to 912. Thus, this model lost only two true positives in exchange for not contacting 131 customers unnecessarily. Paired McNemar's test revealed that XGBoost predicted the correct class more frequently in those samples on which the models disagreed ($p = 5.84 \times 10^{-8}$). From the perspective of the outreach team with constrained resources, it is a more preferable operating point. Logistic Regression can be kept for the sake of transparent governance benchmark

2. Improving the selected model

The next step is decision-threshold selection. Currently, the 0.50 threshold is set because it is technically neutral; however, there should be some business rule. STADIOEquities needs to specify the number of customers that can be contacted, the cost of unnecessary action and the cost of failure to detect likely activators. It will then be possible to find a threshold point maximizing the expected value or fitting the fixed daily capacity by looking at the precision-recall curve. The latter approach will be probably more effective in practice.

Next, XGBoost hyperparameters should be optimized in stratified cross-validation on the training set. These parameters include tree depth, learning rate, number of trees, child weight restrictions, sampling, and scale_pos_weight. The last test period should not be touched till one configuration is selected. Then, probability calibration should be done using a calibration curve and Brier score; Platt scaling and isotonic

regression can be used for that purpose. Calibration will help the client interpret probabilities as real estimated likelihoods and not just rankings.

Finally, class weighting technique should be compared with resampling within training set and cost-sensitive objective function. According to Suguna et al. (2025), an imbalance treatment can greatly affect minority class results, but since their study uses another churn data, SMOTE should be tested against their technique. All candidates should be evaluated based on their PR-AUC, recall, precision, balanced accuracy and cost, not accuracy. Also, feature engineering should be focused on finding early indicators of churn, while interpretation and subgroup analyses can check stability of the model.

3. Adaptation to the SS1 data request

It is necessary to rename the label to first settled deposit within 30 days of registration. Every registration must include an observation time, 24-hour cut-off and 30-day outcome window. Registations without a completed 30-day outcome window at the time of extraction must be either censored or omitted; otherwise recent registrations would be mistakenly labeled as non-activators.

The public set of features must be replaced with the SS1 sources: registration and KYC process, early app session and navigation, acquisition channel, consented marketing contacts, support events and other data that exists until 24-hour cut-off. Features derived after the cut-off, deposit features, balance changes and any field derived from the 30-day outcome window must be excluded. It will follow the same rigorous data curation that is applied to exclude the call duration in the current experiment. Entity keys, event timestamps, duplicated accounts, reversals and transactions failures will need to be defined according to the client data quality agreements.

The validation strategy must also become time-dependent. While the public experiment uses one fixed stratified random split for an unbiased comparison of different techniques, deployment will use training on previous cohorts and testing on newer ones. It will show how the campaigns and the market conditions have changed over time. Preprocessing and model pipeline may stay the same in general, but all imputations, encodings, feature definitions, imbalance weights, thresholds and calibration steps must be fitted on the client data. Prior to the deployment STADIOEquities will also approve data access, retention, security, bias detection, explanation and drift monitoring.

4. Alignment with related literature

The findings are supported by the literature in terms of the direction and realistic scale. Moro, Cortez and Rita (2014) conducted comparison of four models on similar Portuguese banking telemarketing data and provided their best performing Neural Network at ROC-AUC 0.80. The present ROC-AUC findings of 0.802 and 0.809 are comparable to the benchmark set by that study. Additionally, their non-linear model performed better than Logistic Regression, which is consistent with the current choice of XGBoost. Moreover, the rolling window design used in their study is more realistic than the time-based validation proposed for the STADIOEquities samples.

The decision not to replace Logistic Regression with the probit variant is supported by Tanvir, Hossain and Jishan (2024). They demonstrated superiority of Bayesian logit in comparison with Bayesian probit on the same public dataset. The boosting ensemble technique and explicitly applied imbalance treatment, which lead to better precision, F1 and PR-AUC in the current study, have been proven to be beneficial for churn classification by Suguna et al. (2025). However, the numerical findings cannot be compared due to differences in sampling, targets, preprocessing and balancing procedures, and validation approach.

Conclusion

The use of XGBoost is strongly suggested to be used in the next stage since it increases the ranking and precision while keeping the recall and decreasing the workload of the false positives. The experiment conducted publicly demonstrates the viability of the proposed process; however, the value proposition to the customers of STADIOEquities is still unclear. The progression should thus depend on the following conditions:

References

Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22-31. https://doi.org/10.1016/j.dss.2014.03.001.

Moro, S., Rita, P. and Cortez, P. (2014) *Bank Marketing* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5K306.

Suguna, R., Suriya Prakash, J., Aditya Pai, H., Mahesh, T.R., Vinoth Kumar, V. and Yimer, T.E. (2025) 'Mitigating class imbalance in churn prediction with ensemble methods and SMOTE', *Scientific Reports*, 15, article 16256. https://doi.org/10.1038/s41598-025-01031-0.

Tanvir, M.F., Hossain, M.M. and Jishan, M.A. (2024) 'Bayesian Regression for Predicting Subscription to Bank Term Deposits in Direct Marketing Campaigns', *arXiv*, arXiv:2410.21539. Available at: https://arxiv.org/abs/2410.21539 (Accessed: 1 September 2026).